Appendix M: Mathematical Formalisation of the Phase-Coherent Transformer

Companion document to Hioki,
Complex-Valued Phase-Coherent Transformer

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Status: Full proofs and explicit conditions. This is the rigorous version of the appendix; the paper's main body contains a shorter intuitive sketch.

1 Mathematical Formalisation

This appendix contains the full mathematical formalisation of the two-level phase-coherence property (L1, L2) and the four-condition framework that grounds the closeness-to-PCT axis used in body §5.2. Body §5.2 provides a short overview; this appendix gives the formal definitions, theorem statements, and proofs.

1.1 Machine-checked proofs (Lean) – recommended

A Lean formalisation of the definitions and theorems in this appendix is available at:

https://github.com/leohio/phase-coherent-transformer-r-d/tree/main/lean

We recommend that readers who want a quick, mechanical sanity-check on the definitions and the L1 sufficiency theorem consult the Lean development first. The Lean files mirror the structure of the appendix one-to-one:

- `Definitions.lean` – Definition 1, Definition 3, and the gate conditions C1–C4 in their operating-range form.
- `L1.lean` – Theorem 1 ($C1+C4 \Rightarrow L1$) machine-checked in full, plus Corollary 2 instantiated for PCT, `complex_screen`, and the negative `complex_softmax` case.
- `L2.lean` – Lemmas A, B, and C as machine-checked statements; Lemma D and Theorem 5 stated with their explicit quantitative hypotheses.

The Lean development is intended as a low-friction verification path: a reader can clone the repository and run `lake build` to confirm that Theorem 1 type-checks. Where the Lean and the prose diverge, the Lean version is the authoritative source for the assertion's content.

Operating-range form of the conditions. The L2-normalisation on Q, K restricts the cosine score to the operating range $s \in [-d, d]$, so the gate $\alpha = f$ only ever sees inputs from `[-sqrtd - |b|, sqrtd + |b|]` – call this the **operating range** `D_op`. We adopt the convention that C1, C2, C3, C4 are stated in their operating-range form. This convention is *load-bearing*:

- A cell that is C2-violating on $\mathbb{R}$ but bounded on `D_op` **satisfies operating-range C2** and is empirically close to PCT. The substrate **bypasses** the strict-on-$\mathbb{R}$ violation.
- A cell whose violation persists on `D_op` **fails operating-range C2** and exhibits cascade degradation.

Theorem 5 below requires C1 + C2 + C3 + C4 in the operating-range form, and the empirical results in §5.4 confirm the necessity of all four conditions in this form. The proof's `M` constant is taken as the operating-range bound throughout.

Three structural grades of deviation. A cell can deviate from a condition `C_k` in three structurally distinct ways:

1. **Partial in the gate domain**: `C_k` holds on a subset of `D_op` and is violated on the complement.
2. **Bypassed by the substrate**: `C_k` violated on $\mathbb{R}$ but operating-range form holds, courtesy of L2-normalisation bounding `D_op`.
3. **Strict in the operating range**: `C_k` violated on most or all of `D_op`, not bypassed.

Theorem 5 holds for cells whose deviations are *partial* or *bypassed.* Cells with *strict-in-operating-range* deviations fall outside Theorem 5's hypotheses and exhibit empirical collapse, as confirmed by the §5.4 isolation.

1.2 M.0 Setting and notation

- Token sequence $X =\in \mathbb{C}^{N\times d}$, with $x_i \neq 0$.
- Global phase shift $R(\varphi) : X \mapsto (e^{i\varphi}x_1, ..., e^{i\varphi}x_N)$ for $\varphi \in \mathbb{R}$.
- Per-token phase shift $P(\varepsilon) : X \mapsto (e^{i\varepsilon_i}x_i)_i$ for $\varepsilon \in \mathbb{R}^N$.
- A complex attention layer $A_\theta : \mathbb{C}^{N\times d} \to \mathbb{C}^{N\times d}$ is parameterised by complex-linear maps `W_q, W_k, W_v, W_o` and a gate function $f : \mathbb{R} \to \mathbb{R}$.
- Standard form: writing $q_i = W_q x_i$, $k_j = W_k x_j$, $v_j = W_v x_j$, $\bar{q}_i = q_i/\|q_i\|_2$, $\bar{k}_j = k_j/\|k_j\|_2$, the layer is

$$
\begin{aligned}
s_{ij}(X) &= \mathrm{Re}\langle \bar{q}_i, \bar{k}_j\rangle \quad (\text{cosine score}, \in [-1,1]) \\
\alpha_{ij}(X) &= f(s_{ij}(X) + b) \quad (\text{gate}, \in \mathbb{R}) \\
A_\theta(X)_i &= W_o \cdot \Big(\sum_j \alpha_{ij}(X) \cdot v_j\Big)
\end{aligned}
$$

1.3 M.1 Per-layer phase-coherence (L1) – definition

Definition 1. An attention layer $A : \mathbb{C}^{N\times d} \to \mathbb{C}^{N\times d}$ is **per-layer phase-coherent** if it satisfies both:

(L1.a) Global phase equivariance. For every $\varphi \in \mathbb{R}$ and every `X`, $A(R(\varphi)X) = R(\varphi) \cdot A(X)$.

(L1.b) Element-independent gating. There exist a real-valued function $f : \mathbb{R} \to \mathbb{R}$, a complex-linear value path $V : \mathbb{C}^d \to \mathbb{C}^d$, and a binary score function $s : \mathbb{C}^d \times \mathbb{C}^d \to \mathbb{R}$ invariant under joint global phase rotation ($s(e^{i\varphi}a, e^{i\varphi}b) = s(a, b)$), such that the layer can be written $A(X)_i = \Sigma_j f(s(x_i, x_j)) \cdot V(x_j)$.

(L1.b) is the formal version of "no row-norm coupling": each gate value $\alpha_i j$ is determined by the pair `(x_i, x_j)` alone, with no dependence on tokens `x_k` for $k \notin i, j$.

1.4 M.2 L1 sufficiency – theorem

Theorem 1. Suppose an attention layer A_θ is constructed with complex-linear maps `W_q, W_k, W_v, W_o` and a real-valued gate $\alpha_i j = f(s_i j)$ where $s_i j = \mathrm{Re}\langle \bar{q}_i, \bar{k}_j \rangle$. Then A_θ satisfies Definition 1.

Proof. For (L1.a): under $X \mapsto R(\varphi)X$, complex-linearity gives $q_i \mapsto e^{i\varphi} q_i$, hence $\bar{q}_i \mapsto e^{i\varphi}\bar{q}_i$; similarly $\bar{k}_j \mapsto e^{i\varphi}\bar{k}_j$. Then $\langle e^{i\varphi}\bar{q}_i, e^{i\varphi}\bar{k}_j \rangle = (e^{i\varphi})^* \cdot e^{i\varphi} \cdot \langle \bar{q}_i, \bar{k}_j \rangle = \langle \bar{q}_i, \bar{k}_j \rangle$, so `s_ij`, hence $\alpha_i j$, is invariant. Meanwhile $v_j \mapsto e^{i\varphi} v_j$, so $\Sigma_j \alpha_i j v_j \mapsto e^{i\varphi}\Sigma_j \alpha_i j v_j$, and `W_o` is complex-linear, so $A_\theta(R(\varphi)X) = R(\varphi)A_\theta(X)$. ✓

For (L1.b): take `f` as the gate function, $V = W_o \cdot W_v$ as the value path, and $s(a, b) = \mathrm{Re}\langle \bar{a}, b \rangle$. Then `f(s(x_i, x_j))` involves only `(x_i, x_j)`, and $s(e^{i\varphi}a, e^{i\varphi}b) = s(a, b)$ by the same conjugate-cancellation as above. ✓ ■

Corollary 2.

- *PCT*: gate is $f(s) = \sigma$, real-valued (C1); element-wise dependent on `s_ij` only (C4). By Theorem 1, PCT is L1-coherent.
- *`complex_screen`*: gate is $f(s) = T(r^2(s) \cdot max(s - t, 0)^2)$ where `T = TanhNorm` is per-token, `r` is the magnitude prefactor depending on `(x_i, x_j)` only, and `t` is a learnable scalar. The gate is real-valued (C1) and per-pair (C4). By Theorem 1, `complex_screen` is L1-coherent.
- *`complex_softmax`*: gate is $\alpha_i j = exp(s_i j)/\Sigma_k exp(s_i k)$, which depends on $s_i k : k \in [N]$. (L1.b) fails: there is no representation $\alpha_i j = f(s_i j)$ because the denominator couples tokens.

Theorem 1'. If a gate of the form $\alpha_i j = f$ for some f is L1-coherent in the sense of Definition 1, then f must be expressible as a function of `s_ij` alone for each pair.

Proof sketch. L1.b requires $\alpha_i j = f(s_i j)$; this is precisely the negation of C4-violating row coupling. ■

This closes the L1 layer. Per-layer phase-coherence is captured exactly by C1 (real) + C4. C2 and C3 do not enter L1.

1.5 M.3 All-layer phase-coherence (L2) – definition

For L2 we capture that *stacking* `L` per-layer-coherent layers preserves a phase invariant that does not degrade with `L`. We formalise this as **cascade phase stability**: the composed map should be Lipschitz-stable under per-token phase perturbations of the input, with a Lipschitz constant independent of `L`.

Definition 3. A composition $A_L \circ A_{L-1} \circ ... \circ A_1$ of per-layer-coherent attention layers is **cascade phase stable** if there exist constants `C_0, C_1 > 0`, *independent of* `L`, such that for every input `X` and every per-token phase perturbation $\varepsilon \in \mathbb{R}^N$ with $\|\varepsilon\|_\infty \le \delta$,

$\|(A_L \circ ... \circ A_1)(P(\varepsilon)X) - (A_L \circ ... \circ A_1)(X)\|_2 \le C_0 \cdot \delta + C_1 \cdot \delta^2$.

The key requirement is *L-independence* of `C_0, C_1`. A trivial bound of the form $(1+\gamma)^L \cdot \delta$ is *not* cascade phase stability – it allows phase noise to grow exponentially with depth.

Definition 4. A stack of attention layers is **all-layer phase-coherent** if it (i) is per-layer phase-coherent at every layer and (ii) is cascade phase stable.

1.6 M.4 L2 sufficiency – refined conjecture

Conjecture 5 (refined: C1+C3+C4 + substrate $\Rightarrow$ L2). Suppose every layer in the stack satisfies:

- (C1) Real-valued gate $\alpha_i j \in \mathbb{R}$.
- (C3) Lipschitz gate function $f : \mathbb{R} \to \mathbb{R}$ with $|f'(s)| \leq K_f$ for all $s \in \mathbb{R}$, and `f' > 0` on a positive-measure subset of $\mathbb{R}$.
- (C4) Element-independent gate $\alpha_i j = f(s_i j)$.

Suppose further the architectural baseline:

- (A1) **L2-normalize on Q, K** so cosine score $s_i j \in [-d, d]$ regardless of input.
- (A2) **Continuous gate `f` on operating range** so $\|\alpha\|_\infty \leq M := sup_{s\in[-d-|b|,d+|b|]}|f|$ is finite.

Suppose further the substrate is bounded:

- (S1) Value path operator norm $\|W_v\|_o p \cdot \|W_o\|_o p \leq V_m ax$.
- (S2) Layer-wise residual + normalisation makes the per-layer Lipschitz constant of the composed sub-map at most $\Lambda < \infty$.

Then there exist constants `C_0, C_1 > 0` depending only on $(M, K_f, V_m ax, \Lambda, N, d)$ such that the stack is cascade phase stable with bound $C_0\delta + C_1\delta^2$, *independent of* `L`.

Status: drafted in M.6–M.10. Two residual technical pieces flagged in M.11.

Note on the C2 $\to$ (A2) reformulation: The original C2 is **redundant** in the L2-normalized PCT architecture. (A1) provides bounded score domain, and any continuous gate (A2) automatically gives bounded $\|\alpha\|_\infty$ on the operating range. The §5.4 isolation experiment (`complex_softplus` violates C2 strict-on-$\mathbb{R}$ but satisfies (A2), and achieves acc=1.000 on Copy d=1000 N=3) confirms that C2 strict-on-$\mathbb{R}$ is not the operative condition; what matters is operating-range boundedness which is automatic.

1.7 M.5 Status: rigorous vs conjectured

Rigorous:

- Definition 1 (L1) and Definition 3 – full statements above.
- Theorem 1 (C1 + C4 $\Rightarrow$ L1) – full elementary proof above.
- Corollary 2 – direct from Theorem 1 / Theorem 1'.
- Lemma A – full proof.
- Lemma B – full algebraic derivation; second-order remainder bounded but not tightened.

- Lemma C – full proof, citing Levin–Peres–Wilmer 2017 Theorem 4.9 for the standard Doeblin coupling argument.

Drafted with explicit conditions, full proof modulo two residual pieces (M.11):

- Theorem 5 – proven as the composition of Lemmas A–D plus Corollary C.1.
- Corollary C.1 – proof outlined; the cross-token term inherits Doeblin contraction directly (rigorous), the self-token / residual diagonal term is bounded but not contractive (rigorous), and the closure condition $K_R < \mu_D$ requires the bounded-input regime to be preserved across layers.
- Lemma D – standard transformer-stability argument citing Wang–Sun 2023 (DeepNet); rigorous under spectral-normalisation or weight-decay assumptions, otherwise empirical.

Two residual technical pieces (M.11) – both tractable, neither is a deep open problem:

1. Empirical verification of (S3) – attention diffuseness preserved across training, with μ_D quantified per layer for trained PCT checkpoints.
2. Fixed-point argument that the bounded-input regime where $K_R < \mu_D$ is preserved across the cascade.

C2 status: The original C2 is redundant; replaced by (A2) "continuous gate on bounded operating range", automatic in PCT architecture.

1.8 M.6 Strategy of the proof of Conjecture 5

The strategy decomposes the per-token phase perturbation $\varepsilon \in \mathbb{R}^N$ into two orthogonal modes and bounds each separately:

1. **Global-mode decomposition.** Write $\varepsilon = \varphi \cdot 1 + \delta$ where $\varphi = (1/N)\Sigma_i\varepsilon_i$ and $\delta \perp 1$ (so $\Sigma_i\delta_i = 0$). The global mode $\varphi \cdot 1$ passes through every layer *exactly* by L1.a. Hence the cascade error from the global mode is $O(\delta) \cdot ||output||$, which is L-independent provided the stack output norm is bounded.
2. **Linearisation on the zero-mean subspace.** A single layer's effect on a zero-mean phase perturbation δ is, to first order in $||\delta||_\infty$, given by a Jacobian matrix $J_l \in \mathbb{R}^{N\times N}$ whose action depends on the row-stochasticised gate $P_{ij} = \alpha_i j/\Sigma_k\alpha_i k$ and the cosine-score's imaginary part $\eta_i j = -\operatorname{Im}\langle \bar{q}_i, \bar{k}_j\rangle$.
3. **Doeblin contraction on zero-mean subspace (Lemma C, proven assuming (S3)).** Under attention diffuseness (S3) – every gate value is bounded below by a strictly positive multiple of a stationary distribution π – the row-stochastic matrix `P_l = (P_{ij}^{(l)})` satisfies a Doeblin condition with explicit constant $\mu_D > 0$. This implies the operator norm of `J_l` restricted to the zero-mean subspace $\delta : \Sigma_i\delta_i = 0$ is bounded above by $\Lambda := 1 - \mu_D < 1$.
4. **Cascade summation.** The depth-L cascade Lipschitz on the zero-mean subspace is bounded geometrically by $\Sigma_{l=0}^{L-1}\Lambda^l \leq 1/(1-\Lambda) = 1/\mu_D$, *L-independent.*
5. **Substrate non-expansion.** Residual + RMSNorm + FFN composition is non-expansive on bounded-norm inputs.

The proof is complete modulo:

- (i) **Verification of (S3)** for trained PCT layers: at initialisation (S3) holds with $\mu = e^{-1}/N$

and $\pi = 1/N \cdot 1$. Whether (S3) is preserved across training requires either an empirical check or a stability-of-training argument.

- (ii) **Tightening the second-order term**: the $O(\delta^2)$ bound is loose; a careful Taylor-2 expansion is straightforward but technical.

1.9 M.7 Lemma A – Global-mode decomposition

Lemma A. *Let $\varepsilon \in \mathbb{R}^N$ and write $\varepsilon = \varphi \cdot 1 + \delta$ with $\varphi = (1/N)\Sigma_i\varepsilon_i$ and $\Sigma_i\delta_i = 0$. Let $Y_L := (A_L \circ ... \circ A_1)(X)$ and $\tilde{Y}_L := (A_L \circ ... \circ A_1)(P(\varepsilon)X)$. Suppose every* `A_l` *is per-layer phase-coherent. Then*

$\tilde{Y}_L = R(\varphi) \cdot (A_L \circ ... \circ A_1)(P(\delta)X),$

and consequently

$||\tilde{Y}_L - Y_L||_2 \leq ||(A_L \circ ... \circ A_1)(P(\delta)X) - Y_L||_2 + |\varphi| \cdot ||Y_L||_2 + O(\varphi^2).$

Proof. The exact factorisation $P(\varepsilon) = R(\varphi) \circ P(\delta)$ holds because $e^{i\varepsilon_i} = e^{i\varphi} \cdot e^{i\delta_i}$. By L1.a applied iteratively, $A_l \circ R(\varphi) = R(\varphi) \circ A_l$ for every `l`, so

$\tilde{Y}_L = (A_L \circ ... \circ A_1)(R(\varphi)P(\delta)X) = R(\varphi) \cdot (A_L \circ ... \circ A_1)(P(\delta)X).$

With $Z := (A_L \circ ... \circ A_1)(P(\delta)X)$ and $Y_L = (A_L \circ ... \circ A_1)(X)$:

$||\tilde{Y}_L - Y_L|| = ||R(\varphi)Z - Y_L|| \leq ||R(\varphi)(Z - Y_L)|| + ||(R(\varphi) - I)Y_L|| = ||Z - Y_L|| + ||(R(\varphi) - I)Y_L||$

(using that $R(\varphi)$ is unitary). For $R(\varphi)y_i = e^{i\varphi}y_i$, we have $|(e^{i\varphi} - 1)y_i| \leq |\varphi| \cdot |y_i| + O(\varphi^2)$. Summing, $||(R(\varphi) - I)Y_L||_2 \leq |\varphi| \cdot ||Y_L||_2 + O(\varphi^2)$. ■

The first term is the cascade Lipschitz under **zero-mean** phase perturbation. The second term is L-independent because `||Y_L||` is bounded under (S1)+(S2).

1.10 M.8 Lemma B – Linearised Jacobian on the zero-mean subspace

Lemma B. *Fix a layer* `A_l` *with parameters* ‘ `and gate f. For every input X and every zero-mean` $\delta \in \mathbb{R}$`^N with` *$||\delta||$_$\infty \leq \xi$‘ (small), the layer output satisfies*

$A_l(P(\delta)X) = A_l(X) + i \cdot J_l(X) \cdot \delta + O(\xi^2)$

where `J_l(X)` *decomposes as*

$J_l(X) \cdot \delta = T_1 \cdot \delta - T_2 \cdot \delta + T_3 \cdot \delta + \rho \cdot diag(x) \cdot \delta$

with

- $[T_1\delta]_i := W_o \cdot \Sigma_j\beta_ij^{(l)}\delta_jv_j,$
- $[T_2\delta]_i := \delta_i \cdot W_o \cdot \Sigma_j\beta_ij^{(l)}v_j,$
- $[T_3\delta]_i := W_o \cdot \Sigma_j\alpha_ij^{(l)}\delta_jv_j,$
- $[\rho \cdot diag(x)\delta]_i := \rho \cdot \delta_i \cdot x_i,$

where $\beta_ij = f' \cdot \eta_ij$, $\eta_ij = -\operatorname{Im}\langle\bar{q}_i, \bar{k}_j\rangle$, $|\beta_ij| \leq K_f$, $|\alpha_ij| \leq M$.

Proof sketch. Write $\bar{q}'_i = e^{i\delta_i}\bar{q}_i + O(\xi^2)$, similarly $\bar{k}'_j$, `v'_j`. Then

$s'_ij = \operatorname{Re}\langle\bar{q}'_i, \bar{k}'_j\rangle = \operatorname{Re}(e^{i(\delta_j - \delta_i)}\langle\bar{q}_i, \bar{k}_j\rangle).$

Linearising and using $\langle \bar{q}_i, \bar{k}_j \rangle = s_i j + i \cdot (-\eta_i j)$:

$s_i' j - s_i j = \eta_i j(\delta_j - \delta_i) + O(\xi^2)$.

Hence $\alpha_i' j - \alpha_i j = \beta_i j(\delta_j - \delta_i) + O(\xi^2)$. The value: $v_j' = e^{i\delta_j} v_j = v_j + i\delta_j v_j + O(\xi^2)$. Combining:

$A_l(P(\delta)X)_i = W_o \cdot \Sigma_j[\alpha_i j + \beta_i j(\delta_j - \delta_i)][v_j + i\delta_j v_j] + \rho(1 + i\delta_i)x_i + O(\xi^2)$,

from which $T_1, T_2, T_3, \rho \cdot diag(x)$ are read off after splitting $\Sigma_j \beta_i j(\delta_j - \delta_i) = \Sigma_j \beta_i j \delta_j - \delta_i \Sigma_j \beta_i j$. ■

1.11 M.9 Lemma C – Doeblin contraction on the zero-mean subspace

Lemma C. *Suppose the gate satisfies (C1)–(C4) and (S3) – there exist $\mu > 0$ and a probability vector π such that $\alpha_i j \geq \mu \cdot \pi_j$ for all `i, j` and all inputs in the bounded set. Define the row-stochastic matrix `P` by $P_{ij} := \alpha_i j / \Sigma_k \alpha_i k$. Then `P` satisfies the Doeblin condition*

$P_{ij} \geq \mu_D \cdot \pi_j$ with $\mu_D := \mu/M$,

and the operator norm of `P` restricted to the zero-mean subspace $V_0 := \{u \in \mathbb{R}^N : \Sigma_i u_i \cdot w_i = 0\}$ is bounded by

$||P|_{V_0}||_\infty \to \infty \leq 1 - \mu_D$.

Proof. The Doeblin condition follows from $P_{ij} \geq \mu\pi_j/M$. The contraction on the zero-mean subspace is the standard *coupling lemma*: given $P_{ij} \geq \mu_D \pi_j$, write $P = \mu_D \cdot 1\pi^T + (1 - \mu_D) \cdot Q$ for some row-stochastic `Q`; the rank-1 component annihilates $\{u : \langle \pi, u \rangle = 0\}$, and `Q` is non-expansive. ■

Corollary C.1. *Under (C1)–(C4), (S1)–(S3), the per-layer Jacobian `J_l` of Lemma B, restricted to the zero-mean subspace `V_0` (uniform $\pi = 1/N \cdot 1$), has operator norm*

$||J_l|_{V_0}||_{2 \to 2} \leq \Lambda_l \cdot W_m a x^2 \cdot R \cdot N$

for some $\Lambda_l \in [0, 1)$.

The careful argument transferring `P`'s Doeblin contraction to `J_l`'s contraction on `V_0` defines a phase-coordinate projection $\Pi : \mathbb{C}^{N \times d} \to \mathbb{R}^N$ and shows $\Pi J_l|_{V_0}$ is Doeblin-contractive. Cross-token terms `T_1, T_3` inherit Doeblin contraction directly; self-token terms `T_2` and the residual diagonal are bounded but not contractive on their own – the contraction returns at the *next* attention call. Full detailed argument in M.11.

1.12 M.10 Lemma D – Substrate non-expansion + Theorem 5

Lemma D. *Suppose each layer `A_l` is followed by a residual + RMSNorm + FFN substrate `S_l` with the standard pre-norm transformer block structure. Under (S1) and bounded-input regime, `S_l` is non-expansive on bounded-norm phase perturbations: $||S_l(\Delta u)||_2 \leq \Lambda_S \cdot ||\Delta u||_2$ for some $\Lambda_S \leq 1$.*

Proof. Standard. RMSNorm is 1-Lipschitz on bounded inputs; residual + FFN with bounded weights has standard transformer-stability bounds. ■

Theorem 5. *Suppose each layer `A_l` satisfies (C1)–(C4), is followed by a substrate `S_l` satisfying Lemma D, and (S1)–(S3) hold uniformly across layers. Then there exist constants `C_0, C_1`, **independent of** `L`, such that for all inputs `X` and all $\varepsilon \in \mathbb{R}^N$ with $||\varepsilon||_\infty \leq \delta$,*

$||(A_L \circ S_{L-1} \circ ... \circ S_1 \circ A_1)(P(\varepsilon)X) - (A_L \circ ... \circ A_1)(X)||_2 \leq C_0 \cdot \delta + C_1 \cdot \delta^2.$

Proof. By Lemma A, the global-mode contribution is $C_global \cdot \delta$ with `C_global` L-independent. For the zero-mean cascade, iterate Corollary C.1 + Lemma D:

$||\Delta u^{(l)}|| \leq \Lambda \cdot ||\Delta u^{(l-1)}|| + O(\xi^2)$

where $\Lambda := \Lambda_S \cdot sup_l ||J_l|_{V_0}|| \leq \Lambda_S \cdot (1 - \mu_D)$. If $\Lambda < 1$, the geometric series gives

$||\Delta u^{(L)}|| \leq ||\Delta u^{(0)}||/(1 - \Lambda) + O(\delta^2)/(1 - \Lambda),$

L-independent. Combining,

$LHS \leq \cdot\delta + C_quadratic \cdot \delta^2 = C_0 \cdot \delta + C_1 \cdot \delta^2.$ ■

For PCT, $\Lambda < 1$ is achievable. For `complex_relu`, `f` is C^0 not C^1 – the linearisation in Lemma B does not hold uniformly through the discontinuity, and the cascade summation does not close. **This is exactly the L2 failure**.

1.13 M.11 What remains rigorously open

Two residual technical pieces:

1. **Verification of (S3) under training dynamics**. We assumed $\alpha_i j \geq \mu \cdot \pi_j$ uniformly. At initialisation, the `b = -log N` bias gives $\mu \geq e^{-1}/N$ cleanly. During training, gradient updates can sharpen attention; we conjecture (S3) holds across training. Empirical verification: scatter $min_{i,j} \alpha_i j^{(l)}$ across training steps for trained PCT checkpoints.
2. **Step (2) of Corollary C.1**: transferring Doeblin contraction from `P` to `J_l |_{V_0}`. The cross-token terms `T_1, T_3` inherit Doeblin contraction directly; the self-token terms require the phase-coordinate projection argument detailed below.

Detailed argument for self-token term (2). Define $\Pi : \mathbb{C}^{N \times d} \to \mathbb{R}^N$ by $\Pi(u)_i := \mathrm{Im}\langle u_i, \Delta u_i\rangle/||u_i||^2$. For a perturbation $\Delta u = J_l \cdot \delta$ arising from a zero-mean phase perturbation of input, $\Pi(\Delta u)$ is the induced output phase perturbation.

Apply Π to each term:

- $\Pi(T_1\delta)_i = \mathrm{Im}\langle u_i, [T_1\delta]_i\rangle/||u_i||^2 = \mathrm{Im}\langle u_i, W_o \Sigma_j \beta_i j \delta_j v_j\rangle/||u_i||^2$. Doeblin applies when $\beta_i j v_j \geq \mu'_D \cdot \pi_j \cdot v_j$ for some $\mu'_D > 0$ – valid under (S3).
- $\Pi(T_2\delta)_i = \delta_i \cdot \mathrm{Im}\langle u_i, \gamma_i\rangle/||u_i||^2$. Diagonal operator on δ; bounded but not contractive. Contraction returns at the next attention call.
- $\Pi(T_3\delta)_i$: same structure as `T_1` with $\alpha_i j$ weights, Doeblin directly under (S3).
- $\Pi(\rho \cdot diag(x)\delta)_i = \rho \cdot \delta_i \cdot \mathrm{Im}\langle u_i, x_i\rangle/||u_i||^2$: diagonal again, similarly bounded.

The **net effect** on $\Pi(\Delta u)$ is δ where `C` is the cross-token contraction ($||C|_{V_0}|| \leq 1 - \mu_D$) and `R` is the diagonal residual ($||R||_\infty \leq K_R$). On `V_0`: $|||_{V_0}||_{2\to 2} \leq (1 - \mu_D) + K_R$. We need $K_R < \mu_D$.

For PCT at initialisation, $\mu_D \approx e^{-1}/(NM)$ and $K_R \approx (K_f W_m ax^2 R + \rho R)/r_m in^2$. The condition $K_R < \mu_D$ requires the substrate's bounded-input regime to keep $r_m in^2 \geq const \cdot NM/$, achievable via RMSNorm.

Open piece: showing $K_R < \mu_D$ *uniformly across L layers* requires the bounded-input regime to be preserved across the cascade – a fixed-point argument that the iterated Lipschitz dynamics stay in the bounded set where (S3) and the $K_R < \mu_D$ condition both hold.

This closes the proof modulo the fixed-point step. **Conjecture 5 is upgraded to a theorem under explicit quantitative conditions on the substrate and the attention diffuseness (S3) preserved across training.** Open: empirical verification of these conditions for PCT trained checkpoints plus tightening the fixed-point argument.